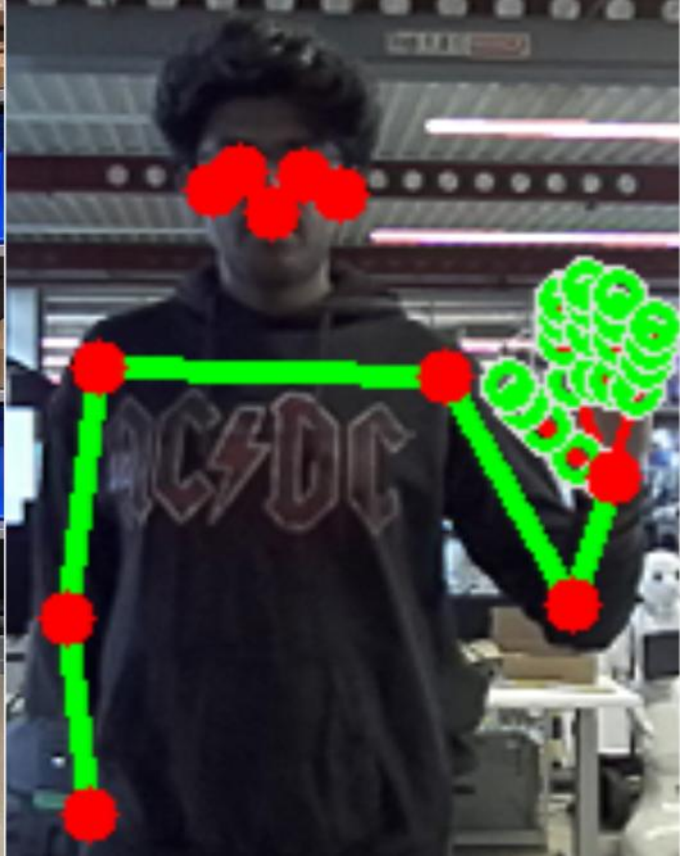

Gesture Recognition for Enhanced Human-Robot Collaboration in Intralogistics Using CNN-LSTM Frameworks

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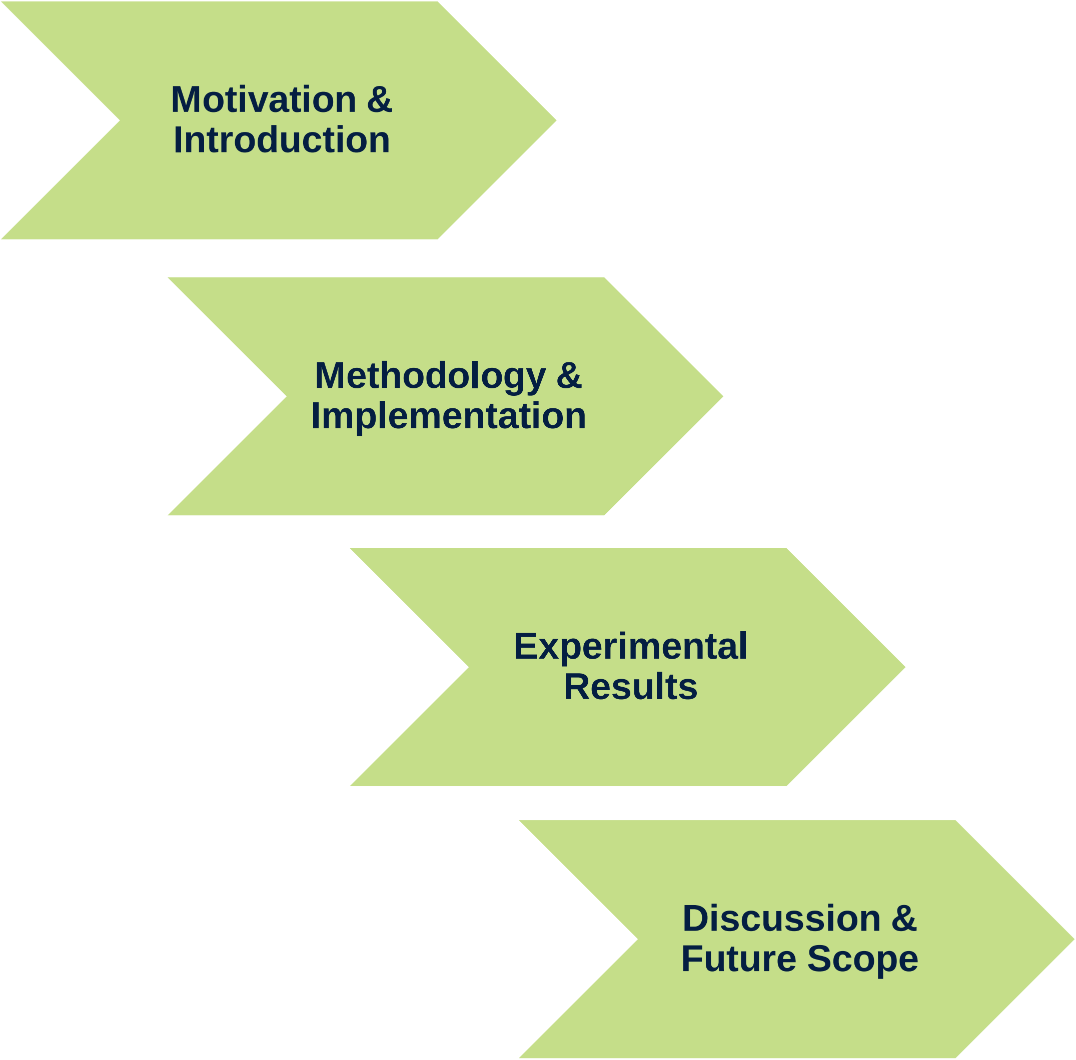
Technische Fakultät

The objective of the thesis is to create a gesture recognition system for enhanced human-robot collaborations in intralogistics scenarios.

Agenda

- **Motivation and Introduction**
  - Need for enhanced human-robot interaction in intralogistics
- **Methodology and Implementation**
  - High-level System Architecture
  - Keypoint Extraction and Preprocessing
  - Custom-defined gesture dataset
  - Functional modes with gestures and its properties
  - Gesture recognition workflow
  - Object detection
- **Experimental Results**
  - Model efficiency via Quantitative Evaluation
  - Performance analysis by Qualitative Evaluation
- **Discussion and Future Scope**
  - Insights into Potential Applications
  - Enhancements for the Proposed Methodology



Introducing a deep learning-based gesture recognition system to enhance intuitive, contactless collaboration with AMRs in dynamic warehouse environments.

FAPS

Motivation

- Manual intralogistics processes are labor-intensive, inefficient, and costly.
- AMRs lack intuitive, non-contact interaction methods.
- Gesture recognition offers a natural alternative to touch/voice input, as a contactless method.
- Vision-based gestures enable safer, intuitive, and hands-free communication.

Introduction

- Built on Linde P50C tow tractor, adapted for autonomous indoor logistics.
- Powered by RTX 4060 GPU, Intel i7, and 16 GB RAM for high-speed inference.
- Proposes a CNN-LSTM gesture recognition system.
- Equipped with ZED 2i for RGB-depth gesture recognition.
- Integrated with ROS 2 for gesture-command mapping.

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Motivation & Introduction

Methodology & Implementation

Experimental Results

Discussion & Future Scope

Suggested vision-based gesture recognition architecture, including static and dynamic gesture for an autonomous mobile robot control in real-world Scenarios.

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Visual Sensing

- ZED 2i stereo camera captures synchronized RGB + depth streams.

Preprocessing

- RGB frames converted to BGR; depth maps aligned.

Keypoint Extraction

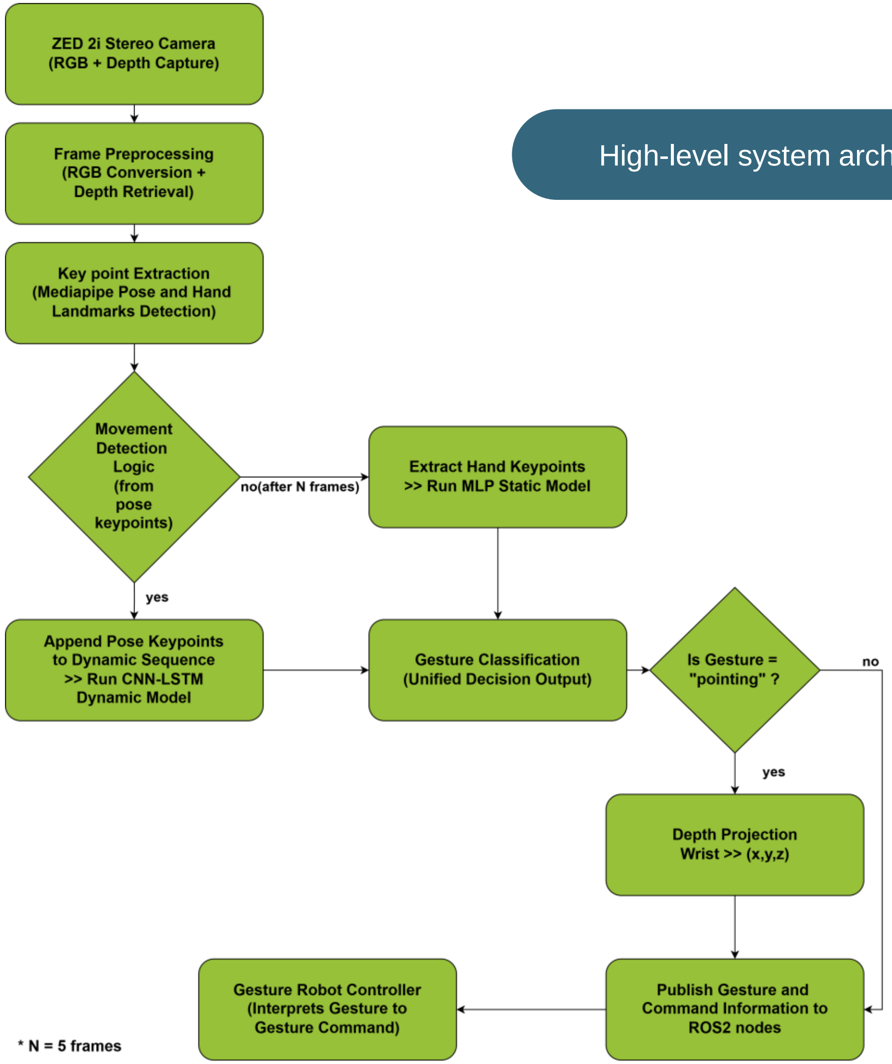
- MediaPipe detects 2D pose and hand landmarks for gesture tracking.

Movement Detection Logic

- Monitors pose changes between frames to detect motion.
- Motion triggers dynamic gesture pipeline; no motion triggers static pipeline. (where N is the no. of frames)

Gesture Classification and Inference

- Static: MLP model on hand landmarks (e.g., open palm).
- Dynamic: CNN-LSTM model on pose sequences (e.g., waving).
- Uses confidence filtering (≥ 0.75).



High-level system architecture

Keypoint extraction and preprocessing for static and dynamic gesture recognition; spatial and temporal modeling of hand and pose landmarks using MediaPipe and CNN-LSTM.

MediaPipe Library and Gesture Preprocessing:

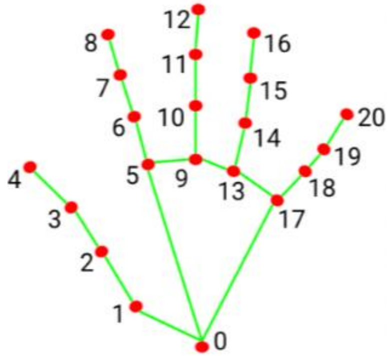
- **MediaPipe** used for real-time pose and hand landmark detection, processes RGB input from ZED 2i Camera.

STATIC

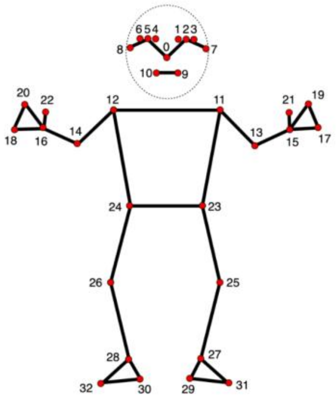
- Detects **21 hand landmarks** (x, y) per frame.
- Converts coordinates to **relative values** using wrist (index 0) as reference.
- Applies **normalization** by dividing all values by the maximum absolute coordinate.
- A compact **42-value feature vector** (21 × 2), stored as rows in a CSV file for training incl. gesture class.

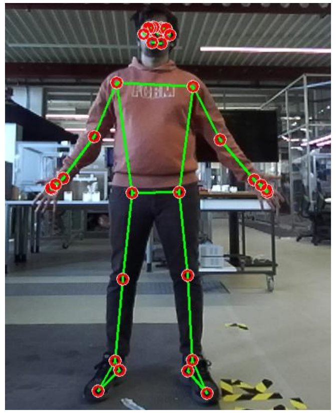
DYNAMIC

- Extract **13 upper-body keypoints** (e.g., nose, eyes, shoulders, wrists).
- Capture **3D (x, y, z)** coordinates per frame → **39 features/frame**.
- Collect **30-frame sequences,** store as .npy files, CNN-LSTM model training.



Hand keypoint landmarks from mediapipe



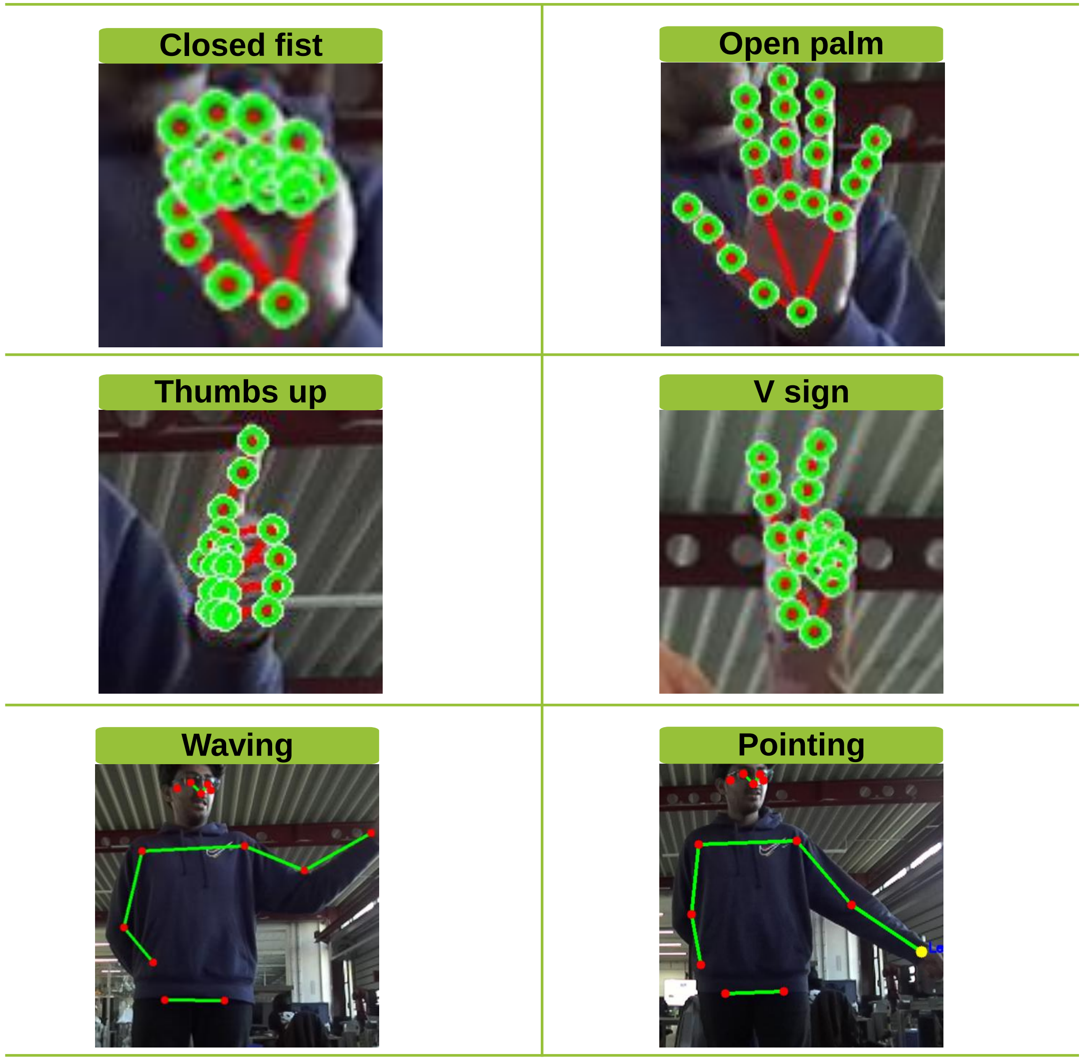


Pose keypoint landmark from mediapipe

Source: Google MediaPipe solution for hand landmarks and pose landmark detection

Custom-defined gesture dataset for human-robot collaboration; including static and dynamic gesture.

- **Six custom defined gestures**
  - Static: Open palm, Closed fist, Thumbs up, and V sign
  - Dynamic: Waving, and Pointing
- **Static gestures**
  - Single-frame input, multi-layer perceptron model, for control/safety (e.g., pause, stop).
- **Dynamic gestures**
  - Multi-frame input, Convolutional Neural Network–Long Short-Term Memory model, for context tasks (e.g., login).



Mapping between functional modes & their corresponding custom-defined gestures, including gesture types, associated commands, and status.

Key Functionalities

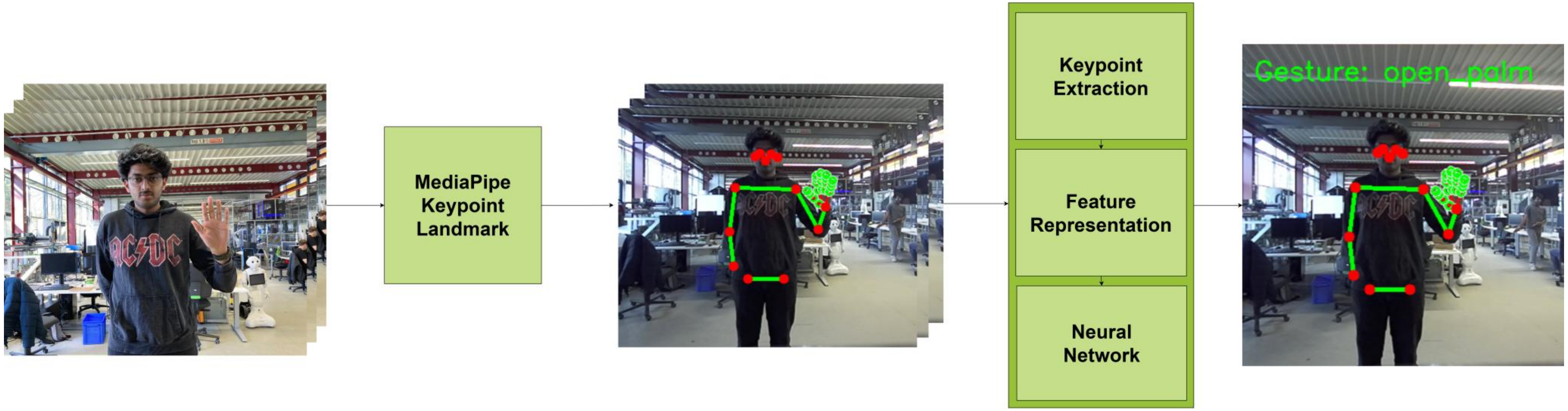
- **Login/Logout mode**
  - Activates/deactivates user interaction with robots.
- **Control Commands**
  - Open palm → Pause / Resume a task
  - Closed fist → Stop task immediately
  - Thumbs up → Mark task as completed
- **Continual learning**
  - V sign → Continual learning without forgetting previously learned task.
- **Navigation**
  - Pointing → Move to indicated location(A) using 3D wrist projection.
- **Following mode**
  - Triggered automatically via person detection (no gesture needed).
  - AMR follows user at a safe distance(~1 m).

Functional mode	Gesture	Type	Command	Status
Login mode	waving	Dynamic	User login	STANDBY
Manoeuvring mode	open_palm	Static	Pause process	paused
	open_palm	Static	Resume process	active
	closed_fist	Static	Stop process	-
	thumbs_up	Static	Task completed	-
	v_sign	Static	-	Learning
Pointing mode	pointing_A	Dynamic	Move to point A	-
Following mode	Object detection	-	Follow user	-
Logout mode	waving	Dynamic	User logout	WAITING

Simplified mapping of gesture to corresponding mode, type, commands, and status

Summing-up of gesture recognition; transforming input frames into classified gestures via landmark detection, keypoint extraction, encoding, and neural network prediction.

- **Camera frames**: Capture video or image sequences as the initial input.
- **Landmark detection**: Process frames to identify and locate key hand and pose landmarks.
- **2D Keypoint extraction**: Extract relevant points (e.g., fingertips, joints) from the detected landmarks.
- **Feature encoding**: Convert keypoints into a structured format (e.g., vectors) for analysis.
- **Gesture classification**: Use a neural network to classify the encoded features into specific gestures.



Enabling person-following mode for enhanced safety using real-time object detection instead of gesture control method.

Autonomous following mode for human-robot navigation

Purpose

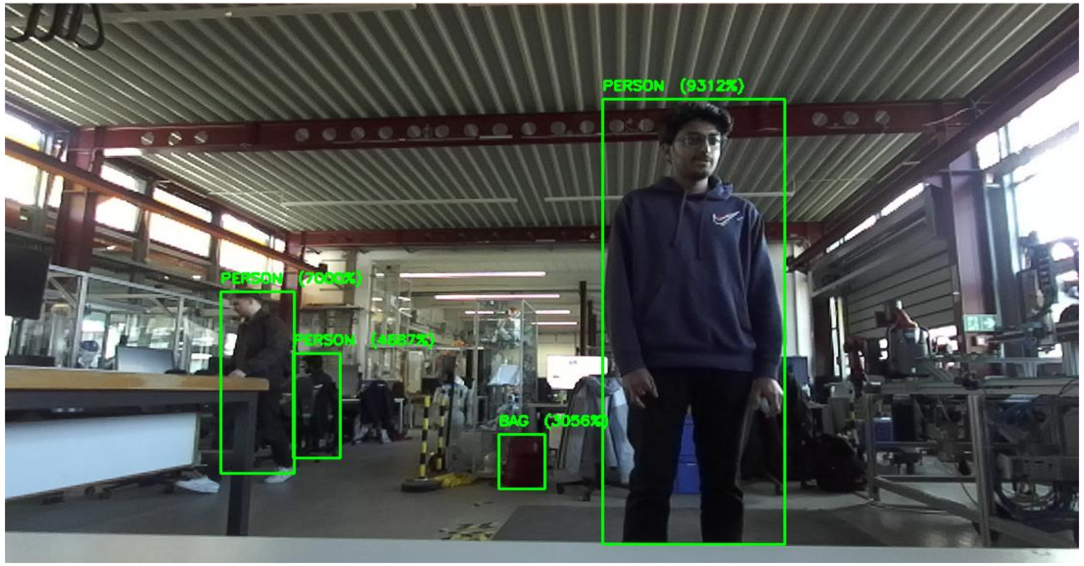
- Enhances safety and intuitiveness in close-proximity interaction using real-time 3D object detection.

Technology

- Powered by ZED 2i stereo camera and ROS 2, bypassing gesture recognition for robust tracking.

Key Features

- Tracks users autonomously while maintaining a safety distance.
- Uses ZED SDK's **MULTI CLASS BOX MEDIUM** model for multi-object detection (e.g., "PERSON").
- Falls back to a safe stop if the user is close or leaves the field of view.



Advantages

- Low-latency processing with positional tracking for seamless following.
- Supports multiple object classes, making it versatile for general-purpose use.
- Visual feedback via bounding boxes and confidence scores.

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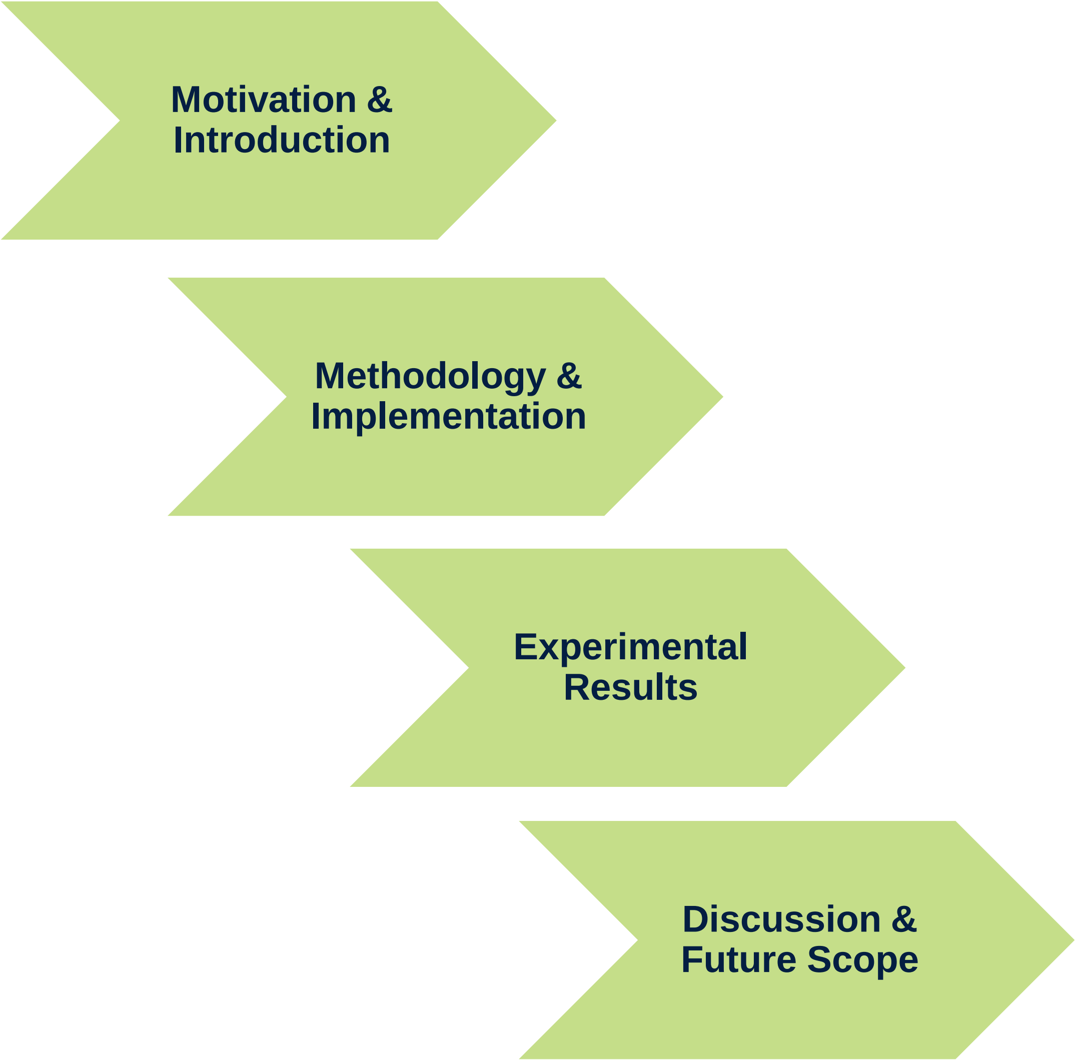
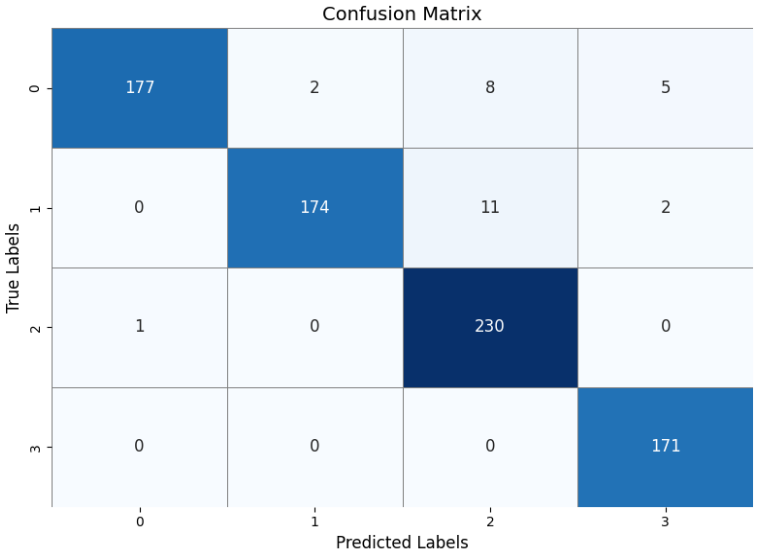


Quantitative evaluation of model performance metrics by analysing accuracy, precision, and recall from Confusion Matrix data.

Static Gesture Model (MLP) Evaluation:

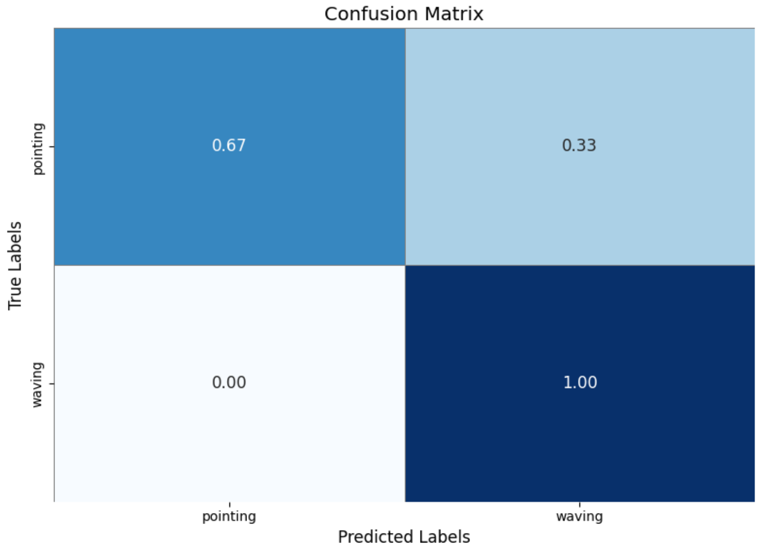
- High Accuracy: Achieved 96.7% test accuracy on four gesture classes.
- Strong Performance: Near perfect classification for closed fist and open palm gestures.
- Minor confusion between “closed fist“ and "thumbs up", likely due to 2D projection similarities.
- Robust Metrics: High precision, recall, and F1-scores across all classes.



MLP Model Evaluation

Dynamic Gesture Model (CNN-LSTM) Evaluation:

- Perfect Detection: 100% accuracy for "waving" gestures.
- Partial Confusion:
  - "Pointing" gestures had 67% accuracy, with 33% misclassified as waving.
  - Likely due to overlapping arm motion patterns and subtle temporal differences.
- Key Insight: Temporal modeling works well for expressive motions but may need refinement for finer gestures.



CNN-LSTM Model Evaluation

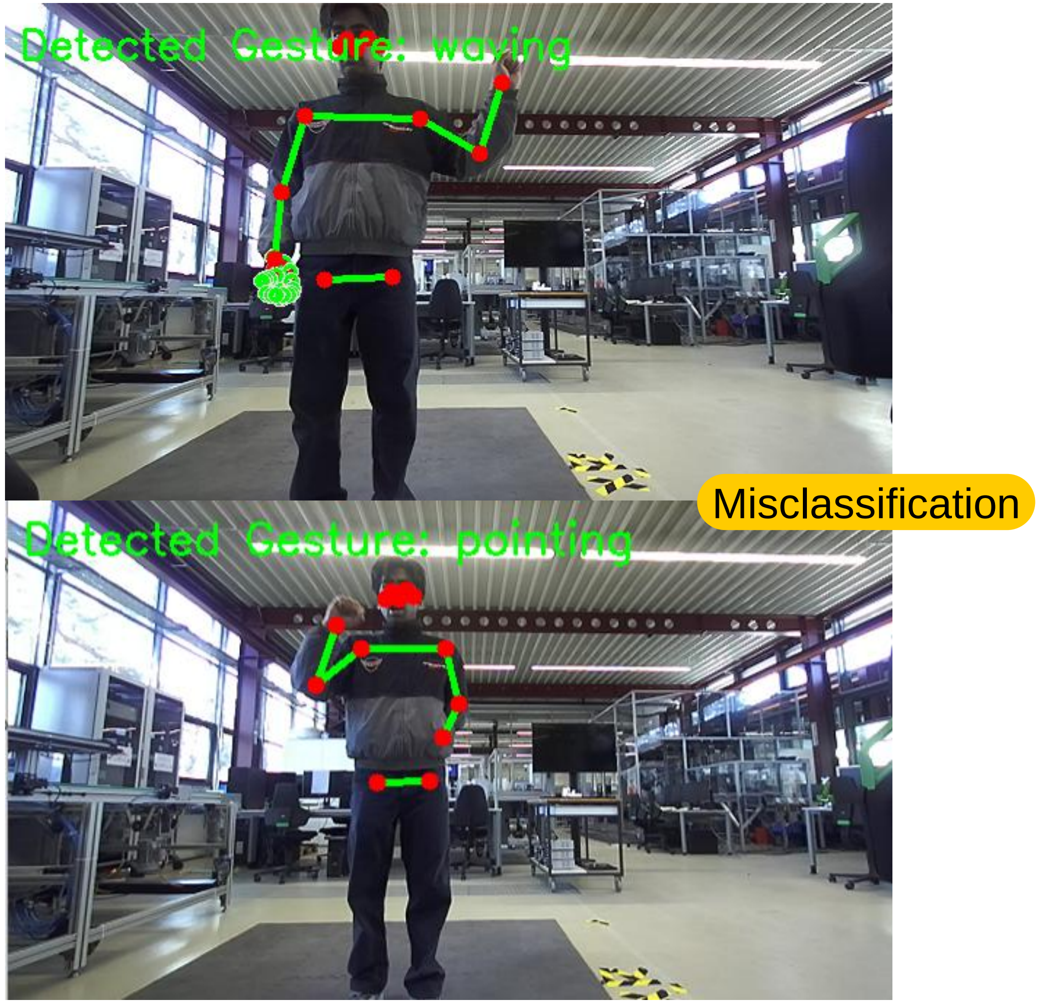
Qualitative analysis by evaluating real-world gesture performance and tracking across different users, gestures, and lightning conditions.

Qualitative Analysis

- Tested with 2 participants at a fixed distance of ~1 meter from the ZED 2i camera.
- Robust classification for 4 static and 2 dynamic gestures across different users and lighting.
- Both models generalized well to individual variations in hand size, posture, and movement.
- System self-switches between static and dynamic modes based on movement detection (set to 5 fps).

Misclassification

- Since dynamic gestures trained only on left-hand sequences with limited samples (30 sequence per class).
- Model switching delay causes in-correct predictions.
- Model struggles with right-hand gestures and unseen poses due to limited training data.
- Common confusions between “waving” and “pointing” under ambiguous motion.



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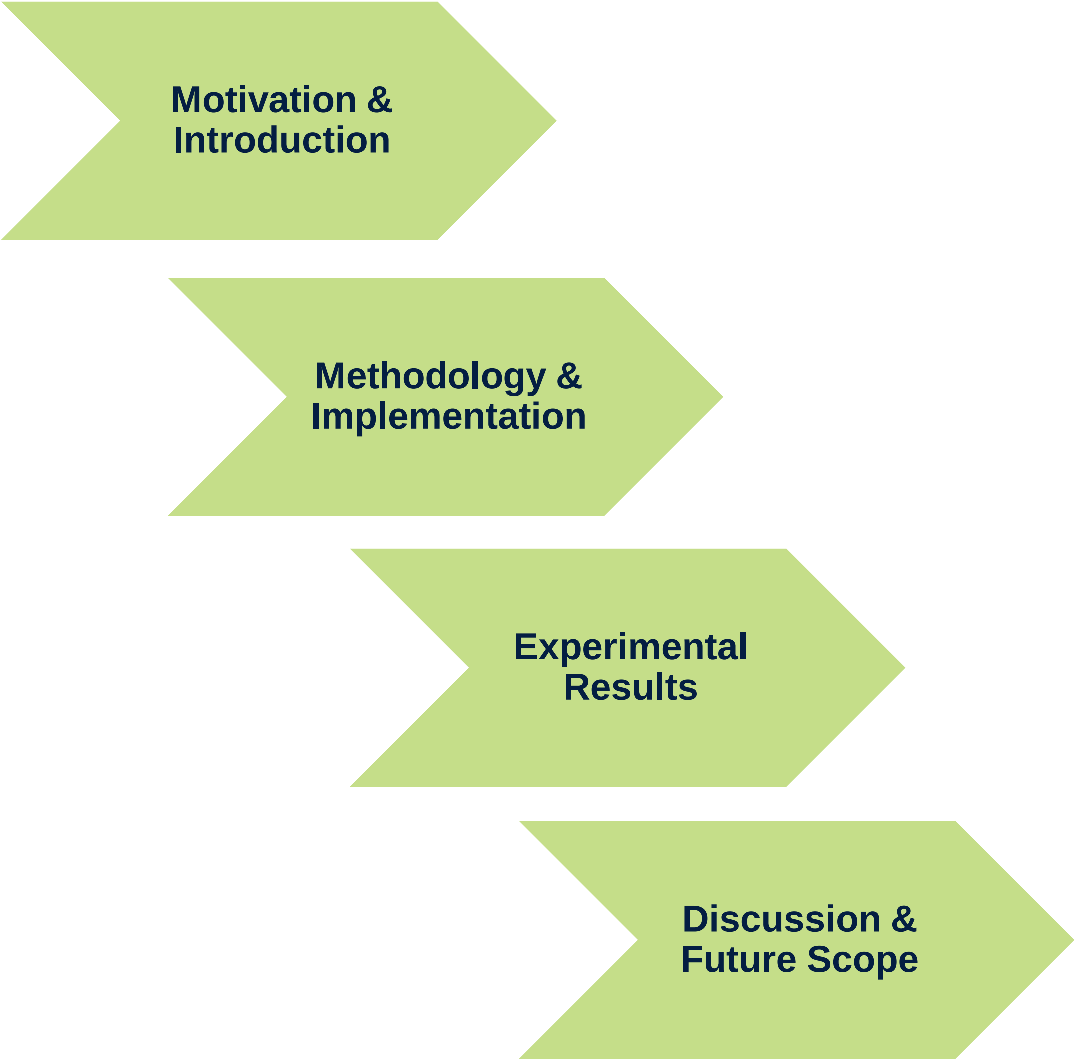

This thesis successfully designed and deployed a gesture recognition system enabling real-time, robust static and dynamic gesture detection for human-robot collaboration.

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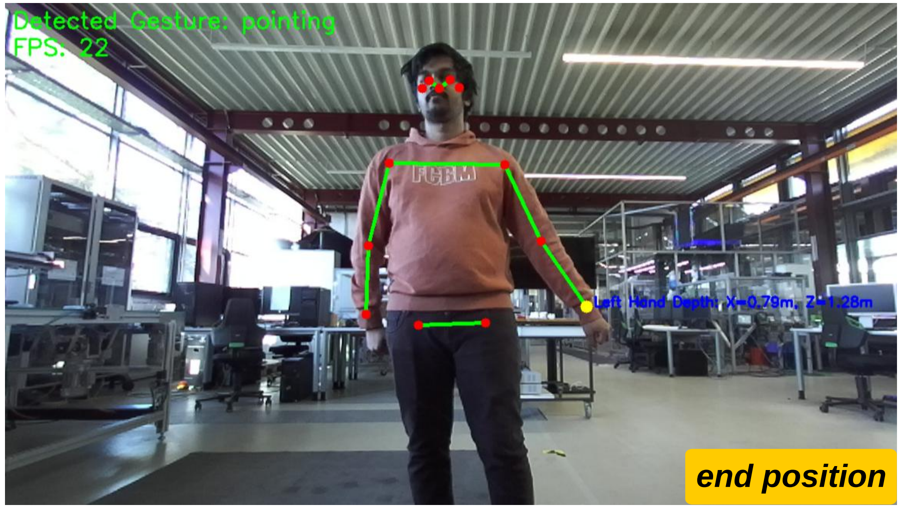
Key Achievements :

- Real-time gesture recognition system using hybrid MLP and CNN-LSTM model.
- Recognizes static (2D keypoints) and dynamic (3D posture) gestures.
- Achieved 96.7% (static), 100% (waving), and 67% (pointing) accuracy.
- Integrated with ROS 2 and ZED 2i for real-time operation and control.

Future Scope :

- Extend the training dataset with more gesture sequences, varied user inputs, varied camera perspectives, different hand positions, and under multiple lighting conditions to improve model generalization and accuracy.
- Improve real-time performance by incorporating acceleration frameworks like TensorRT for faster inference.
- Incorporate person detection and tracking alongside gesture recognition to allow user-specific interactions, multi-user support, and better contextual awareness.



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THANK YOU